

MONT-DE-MARSAN AB, France

WMO: 076070

Lat: 43.9097N

Lon: 0.5003W

Elev: 203

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	26.1	18.7	14.2	28.6	22.3	16.9	30.7	21.1	51.1	18.3	51.3	2.5	80	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	92.1	71.6	88.4	70.0	85.2	68.7	73.7	87.1	72.0	84.7	70.4	81.8	6.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.7	110.1	79.5	68.1	103.9	76.6	66.6	98.7	75.1	37.3	87.0	35.8	84.8	34.4	82.1	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	14.7	12.3	17.5	99.2	4.1	3.4	14.6	101.6	12.2	103.6	9.9	105.5	7.0	108.0
			17.1	77.1	4.1	2.0	14.1	78.5	11.7	79.7	9.4	80.8	6.4	82.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	43.3	44.5	50.0	54.4	60.6	66.9	70.3	70.3	64.5	58.6	48.8	44.1	(6)
(7)		DBStd	11.67	6.79	6.70	6.19	5.95	6.21	6.38	5.68	5.81	5.83	6.64	6.54	7.26	(7)
(8)		HDD50	809	222	171	77	23	3	0	0	0	13	97	203		(8)
(9)		HDD65	3709	673	574	465	321	165	53	17	16	78	213	486	648	(9)
(10)		CDD50	3153	14	17	77	154	331	506	628	629	436	280	61	20	(10)
(11)		CDD65	577	0	0	0	2	28	108	180	179	64	16	0	0	(11)
(12)		CDH74	6047	0	1	15	101	401	1167	1742	1752	726	141	1	0	(12)
(13)		CDH80	2223	0	0	0	15	108	442	690	717	226	25	0	0	(13)
(14)	Wind	WSAvg	5.3	5.1	5.4	5.9	6.2	6.0	5.8	5.7	5.1	4.5	4.5	4.7	4.7	(14)
(15)	Precipitation	PrecAvg	38.1	3.8	2.9	2.8	3.7	3.2	2.7	2.2	2.4	3.0	3.5	4.8	3.3	(15)
(16)		PrecMax	48.4	8.7	5.4	5.9	9.1	7.7	6.9	5.0	6.9	6.0	7.6	12.3	6.8	(16)
(17)		PrecMin	30.1	0.6	0.3	0.4	1.0	1.1	0.4	0.5	0.9	1.1	1.0	0.2		(17)
(18)		PrecStd	5.0	2.2	1.6	1.6	2.0	1.4	1.5	1.1	1.5	1.4	1.9	2.7	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.2	69.1	76.0	81.9	88.1	95.6	95.9	99.2	90.5	83.5	70.3	64.3	(19)
(20)		0.4%	MCWB	54.7	54.6	57.3	62.3	67.9	72.9	72.4	72.7	69.1	66.2	60.2	54.7	(20)
(21)		2%	DB	59.0	63.4	71.0	76.8	82.8	89.1	91.2	92.2	85.9	77.8	65.8	59.8	(21)
(22)		2%	MCWB	52.8	52.7	56.1	59.6	65.2	70.0	71.4	72.1	68.0	65.1	57.8	53.4	(22)
(23)		5%	DB	55.9	59.2	66.7	72.1	78.4	85.1	87.5	87.4	82.2	73.8	61.8	57.0	(23)
(24)		5%	MCWB	51.8	51.0	53.8	57.4	63.6	68.6	70.0	70.4	66.5	63.2	56.5	52.8	(24)
(25)		10%	DB	53.6	55.5	62.6	67.6	73.9	80.9	83.6	83.3	78.0	70.0	58.9	54.6	(25)
(26)		10%	MCWB	50.4	50.1	52.3	55.4	61.5	67.0	68.9	69.0	64.9	61.8	54.9	51.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.6	56.5	59.3	64.0	69.7	75.1	75.6	76.2	72.8	68.6	62.1	57.2	(27)
(28)		0.4%	MCDWB	59.5	64.4	70.4	77.7	84.0	88.8	89.2	90.8	85.0	77.5	67.1	60.3	(28)
(29)		2%	WB	54.4	54.1	57.4	61.2	66.9	72.2	73.2	73.9	70.0	66.6	59.4	55.1	(29)
(30)		2%	MCDWB	57.5	59.8	67.4	73.5	80.1	85.8	87.4	86.7	80.8	74.5	63.6	58.3	(30)
(31)		5%	WB	52.6	52.5	55.5	58.8	64.7	70.1	71.5	72.1	68.1	64.7	57.3	53.4	(31)
(32)		5%	MCDWB	55.2	57.3	64.1	68.8	75.6	82.3	84.7	83.9	77.9	71.3	60.7	56.1	(32)
(33)		10%	WB	50.7	50.9	53.7	56.7	62.7	68.0	69.9	70.3	66.4	63.0	55.3	51.7	(33)
(34)		10%	MCDWB	53.1	55.0	60.5	64.9	71.8	78.8	80.7	80.5	75.3	68.6	58.4	54.3	(34)
(35)	Mean Daily Temperature Range		MDBR	15.5	19.1	21.5	21.5	21.3	21.8	22.0	23.0	23.6	20.8	16.6	15.9	(35)
(36)		5% DB	MCDBR	19.0	26.7	30.0	30.7	29.5	29.9	29.8	31.2	31.0	26.2	21.0	19.0	(36)
(37)		5% DB	MCWBR	13.3	16.7	16.0	15.3	13.9	13.4	12.2	12.9	14.2	13.9	13.9	13.0	(37)
(38)		5% WB	MCDBR	15.7	21.8	24.3	26.0	25.6	26.2	25.8	26.0	25.6	21.7	18.2	15.6	(38)
(39)		5% WB	MCWBR	11.8	14.5	13.6	13.4	12.8	12.8	11.2	11.4	12.5	12.0	12.9	11.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.325	0.340	0.373	0.399	0.407	0.420	0.411	0.409	0.382	0.365	0.341	0.322	(40)	
(41)		taud	2.489	2.428	2.342	2.275	2.304	2.300	2.327	2.343	2.404	2.457	2.472	2.488	(41)	
(42)		Ebn at Noon	254	270	273	274	275	271	272	268	267	257	245	243	(42)	
(43)		Edh at Noon	23	28	35	40	40	41	39	37	32	27	23	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	451	742	1115	1427	1657	1863	1854	1658	1352	882	533	419	(44)	
(45)		RadStd	43	89	128	153	138	134	132	113	74	72	62	48	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page